

Open-Set Individual Bat Face Recognition: Identity-Disjoint Evaluation, Pair vs. Embedding Architectures, and the Small-Identity Generalisation Wall

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Abstract

We present an open-set individual face recognition system for fruit bats, designed to support longitudinal behavioural studies where every animal must be identified without prior appearance in the training set. Our primary methodological contribution is a rigorous *identity-disjoint* evaluation protocol: train, validation, and test splits are partitioned at the level of individual identities so that no bat ever appears in more than one split. We show that the widely-used alternative—random pair selection—inflates performance dramatically: a Siamese network that achieves $F_1 = 0.99$ under the random-pair protocol drops to $F_1 = 0.71$ under identity-disjoint evaluation on the same images.

We compare three deep-learning model families—a pair-based Siamese convolutional network and two embedding-based angular-margin architectures (ArcFace and AdaFace built on a ResNet-50 backbone)—across a factorial design covering two fruit-bat species (rousettus and mauritius), three background conditions (green screen, original, and random compositing), and small identity counts (≤ 13 training individuals). The Siamese network is the only family that verifies significantly across *all* conditions (pairwise-head ROC-AUC 0.67–0.86; cosine ROC-AUC 0.82–0.96). The embedding models, by contrast, are governed by the background: on random-composited backgrounds they regress toward chance (rousettus ArcFace 0.665, AdaFace 0.434; not statistically significant), but on coherent green-screen backgrounds they recover sharply and can *surpass* the Siamese network—AdaFace reaches ROC-AUC 0.902 with Top-1 0.916 on mauritius green, at the same small identity count. We therefore characterise the embedding-model shortfall as a *background-sensitivity* effect rather than a fixed small-identity ceiling: with stable backgrounds, angular-margin embeddings transfer to unseen identities even at this scale, whereas random-background nuisance variation—not identity count alone—is what collapses them.

All experiments are fully reproducible via a single-command pipeline and are tracked in MLflow.

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1 Introduction

Fruit bats play pivotal roles in tropical and subtropical ecosystems as pollinators and seed dispersers, yet the individual-level behaviour of colony members remains poorly understood. Long-term studies of social dynamics, foraging ranges, and roosting fidelity require reliable identification of individual animals across repeated observations—a task traditionally performed by physical tagging or manual expert annotation, both of which are costly, disruptive to the animals, and difficult to scale.

Automated re-identification from facial imagery offers a non-invasive alternative. The bat face is sufficiently individuated through pelage patterns, vein geometry, and facial proportions to support individual recognition [15]. Advances in deep face recognition for humans [3, 8, 13, 18] provide powerful transfer-learning priors, yet animal datasets pose three challenges absent in standard human benchmarks: (i) *small identity count*—studies typically monitor tens, not millions, of individuals; (ii) *open-set inference*—a deployed system must handle bats not seen during training; and (iii) *environmental variation*—imaging conditions change across sessions, with different backgrounds, lighting, and partial occlusions.

The leakage problem

A critical but often underappreciated methodological issue in animal re-identification research is *identity leakage*: when images are split randomly into train/val/test at the image level rather than the identity level, the same individual can appear in all three partitions. The model therefore sees every identity during training and the test protocol measures closed-set retrieval rather than genuine open-set generalisation. We demonstrate that a Siamese network trained and evaluated under the random-pair protocol achieves $F_1 = 0.99$, whereas the same network evaluated under a strict identity-disjoint protocol on the same images yields $F_1 = 0.71$. The lower number is the scientifically correct one.

Contributions

This work makes the following contributions:

1. We introduce and validate an **identity-disjoint evaluation framework** for bat face recognition, enforced at the manifest level with cryptographic integrity checking, and demonstrate that it is the correct protocol for open-set individual recognition.
2. We conduct a **systematic multi-factor comparison** of pair-based and embedding-based deep recognition architectures across two bat species, three background conditions, and controlled identity counts, using real video data from tracking studies.
3. We show that, at the scale of ecological datasets (≤ 13 training identities), the pair-based Siamese network is the only family that verifies significantly across all conditions, while corrected angular-margin embeddings (ArcFace/AdaFace) match or surpass it on coherent backgrounds—characterising the embedding-model shortfall as a **background-sensitivity** effect (collapse on random-composited backgrounds) rather than a fixed small-identity ceiling.

4. We release a **reproducible 13-package pipeline** (`bat-cli`) that automates the full workflow from raw video to PDF experimental report, with Hydra configuration, MLflow tracking, and Optuna hyperparameter search.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the datasets and the identity-disjoint splitter. Section 4 details the three model families. Section 5 formalises the evaluation protocol and metrics. Section 6 presents experimental results. Section 7 analyses the findings and limitations. Section 8 concludes.

2 Related Work

2.1 Animal re-identification

Automated individual identification from imagery has been explored for a range of species [17]. Early work on sea turtles, whale sharks, and manta rays [12, 15] used hand-crafted texture descriptors. More recently, deep convolutional networks have been adapted from human face recognition to animal identification, typically via fine-tuning on species-specific datasets. A persistent gap in many published systems is the use of identity-leaky evaluation splits, which makes cross-paper comparisons unreliable and overstates deployment-ready performance. Our work explicitly addresses this gap.

2.2 Metric learning and face verification

The Siamese network, originally proposed for signature verification [2] and popularised for one-shot classification [9], learns an embedding space where pairs of the same identity are pulled together and different-identity pairs are pushed apart. The similarity score is computed directly from the pair, which aligns the training objective with the verification task. FaceNet [18] extended this idea with triplet mining on large-scale human face datasets, achieving strong open-set performance.

2.3 Angular margin softmax losses

A parallel line of work treats face recognition as a closed-set classification problem at training time and uses the resulting embedding space for open-set retrieval at test time. SphereFace [11] introduced a multiplicative angular margin; CosFace [19] replaced it with an additive cosine margin; ArcFace [3] used an additive angular margin directly in the angle space, giving sharper decision boundaries and better calibration. These losses share a common structure: the logit for the ground-truth class is modified to require a larger cosine distance from the nearest other class, promoting compact, well-separated clusters in the hyperspherical embedding space.

Sub-center ArcFace [4] relaxes the assumption of a single clean prototype per class, allowing multiple sub-centres to handle noisy labels. AdaFace [8] further adapts the margin magnitude to the estimated quality of each image, using the *feature norm* as a proxy: low-quality images (whose embeddings have smaller norms) receive a smaller margin, reducing their influence on the decision boundary.

While these angular-margin methods dominate public human face benchmarks (where training sets contain millions of identities), their behaviour under very few training identities—as in

ecological datasets—has not been systematically studied. Our experiments fill this gap.

2.4 Biometric evaluation metrics

Standard face recognition benchmarks report rank-1 identification accuracy and verification equal-error rate (EER). For security-critical biometric applications the operating metric is *TAR at a fixed FAR* (true acceptance rate at false acceptance rate), reflecting real-world deployment constraints. In our setting, where test partitions contain at most a few hundred image pairs, FAR targets such as 10^{-3} are effectively degenerate (budgeting fewer than one false match), so we adopt $\text{Recall@FAR} \leq 10^{-2}$ as the headline biometric metric alongside ROC-AUC, which provides a threshold-independent summary of discriminative power. The Youden index [21], $J = \text{TPR} - \text{FPR}$, is used for optimal threshold selection.

3 Dataset

3.1 Species and capture conditions

We work with two fruit-bat species from active research colonies:

- **rousettus (Egyptian fruit bat)**: 12 individuals, 1 059 images. Images were extracted from close-range video recordings; each frame was independently processed by the preprocessing pipeline. All images were captured with randomised background compositing.
- **mauritus (Mauritian flying fox, *Pteropus niger*)**: three background conditions are available for this species, each captured/curated independently and therefore differing in identity and image counts: *random* (composited random textures; 11 individuals, 1 119 images), *green screen* (chroma-key replacement; 19 individuals, 1 598 images), and *original* (natural enclosure background; 17 individuals, 1 038 images). Together they provide a controlled study of background-invariance (see Table 1 for per-split counts).

3.2 Preprocessing pipeline

Raw video frames undergo a five-stage preprocessing pipeline before storage. First, a YOLO-v8 pose estimator (custom weights `face_pose.pt`) detects facial keypoints; the face is then rotated so that the eyes are horizontally aligned, providing pose-normalised crops. A second YOLO-v8 model (`face_seg.pt`) segments the bat face region; the bounding box of the predicted mask drives a square crop with a configurable outer margin (default 10%). Optional background replacement composites one of several procedurally generated backgrounds (blur, gradient, random noise, or downloaded textures). Finally, images are resized to the model’s input resolution and normalised to $[0, 1]$.

Image *quality* is quantified per frame as the variance of the Laplacian $\sigma_{\Delta}^2 = \text{Var}(\Delta I)$, a standard focus measure; this score is stored in the manifest and can be used to filter low-quality frames during dataset construction.

3.3 Manifest structure

All dataset metadata is stored in a *manifest*: a CSV file with the following columns: `path`, `identity`, `species`, `background`, `source`, `augmented`, `split`, `quality`. Each manifest is accom-



Figure 1: Splitting strategies. (a) The random-pair protocol assigns images without regard to identity: every individual (A, B, C) contributes images to all three partitions. (b) The identity-disjoint protocol assigns each individual to exactly one partition, so test identities (D) are never seen during training.

panied by a SHA-256 hash sidecar; loading a manifest with a mismatched hash raises an error, preventing silent data-version drift.

3.4 Identity-disjoint splitting

The central methodological contribution of this work is the enforcement of *identity-disjoint* splits. Given a set of N unique individual identities, the splitter assigns each identity to exactly one of $\{\text{train, val, test}\}$ before any images are allocated. This guarantees that the model never observes images of a test identity during training or validation—the protocol required for honest open-set evaluation.

Formally, let $\mathcal{I}$ be the set of identities. The splitter draws $(n_{\text{val}}, n_{\text{test}})$ identities as $n_{\text{val}} = \lfloor |\mathcal{I}| \cdot r_{\text{val}} \rfloor$, $n_{\text{test}} = \lfloor |\mathcal{I}| \cdot r_{\text{test}} \rfloor$, with the remainder assigned to training. The shuffle is seeded (seed 42) over an alphabetically sorted identity list, making the split fully deterministic.

Figure 1 contrasts the random-pair protocol with the identity-disjoint protocol.

3.5 Dataset statistics

Table 1 summarises the final partition statistics.

Table 1: Dataset split statistics under the identity-disjoint protocol (seed 42). Whole identities are assigned to exactly one split; every condition reserves 3 identities for validation and 3 for test, with the remainder used for training. BG = background condition.

Species	BG	Total imgs	Identities total	Train		Val		Test imgs
				ids	imgs	ids	imgs	
rousettus	random	1 059	12	6	486	3	343	230
rousettus	green	1 093	12	6	490	3	359	244
mauritus	random	1 119	11	5	400	3	343	376
mauritus	green	1 598	19	13	1 229	3	230	139
mauritus	original	1 038	17	11	675	3	175	188

The small identity counts—fewer than twelve training identities in all conditions—are representative of typical ecological monitoring studies and define the challenge regime analysed in Section 6.

4 Methods

We compare two families of face recognition models distinguished by their training objective: *pair-based* models that learn directly from same/different pairs, and *embedding-based* models that learn a compact hyperspherical representation via closed-set classification with an angular margin. Both families share a common inference interface: given two images, cosine similarity of L2-normalised embeddings is computed and compared to a learned threshold.

4.1 Pair-based: Siamese Convolutional Network

The Siamese network [2, 9] consists of two weight-tied embedding towers and a pairwise decision head.

Embedding tower. The tower processes 105×105 RGB images through four convolutional stages with “valid” (zero-padding-free) convolutions:

1. Conv $64@10 \times 10$, ReLU $\rightarrow$ MaxPool 2×2 (stride 2): $105 \rightarrow 96 \rightarrow 48$
2. Conv $128@7 \times 7$, ReLU $\rightarrow$ MaxPool 2×2 : $48 \rightarrow 42 \rightarrow 21$
3. Conv $128@4 \times 4$, ReLU $\rightarrow$ MaxPool 2×2 : $21 \rightarrow 18 \rightarrow 9$
4. Conv $256@4 \times 4$, ReLU: $9 \rightarrow 6$
5. Flatten ($256 \cdot 6 \cdot 6 = 9\,216$) $\rightarrow$ Linear($9\,216, 4\,096$) $\rightarrow$ Sigmoid

Producing a 4 096-dimensional embedding with values in $[0, 1]$.

Pair head. The absolute L1 distance between the two embeddings, $|\mathbf{e}_a - \mathbf{e}_b|$, is passed through a final linear layer ($4\,096 \rightarrow 1$) followed by a sigmoid, producing a similarity score $s \in [0, 1]$. L2 regularisation with weight 10^{-4} is applied via the optimiser, replicating the Keras `12(1e-4)` behaviour.

Loss. Siamese networks are trained with the Focal loss [10]: $\text{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log p_t$, with $\alpha = 0.75$, $\gamma = 2.0$. Focal loss down-weights easy negatives that dominate pairwise training sets, improving calibration of the decision boundary.

Inference. At inference time only the embedding tower is used; the trained pair-head is discarded. Verification scores are computed as the cosine similarity of L2-normalised embeddings, providing a consistent metric space across all three model families.

Figure 2 (left) illustrates the Siamese architecture.

4.2 Embedding-based: ArcFace and AdaFace

Both embedding models share a common backbone and projection block, differing only in their classification head. The architecture maps 112×112 RGB images to a 512-dimensional hyperspherical embedding.

Backbone. A ResNet-50 [6] initialised with ImageNet1K-V2 weights serves as the feature extractor. The original fully connected classification head is replaced by `nn.Identity()`, yielding a 2048-dimensional feature vector. Because the backbone is ImageNet-pretrained, inputs are normalised with the ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) consistently across training, validation, test, and the embedding projections used for visualisation; the from-scratch Siamese tower instead consumes images scaled to $[0, 1]$. We note one deliberate deviation from the cited angular-margin literature: the original ArcFace/AdaFace works use an IR/IR-SE stem (a 3×3 stride-1 convolution preserving spatial resolution) with an Option-E embedding head that flattens the final feature map. We instead use the torchvision ResNet-50 stem (7×7 stride-2 convolution followed by max-pooling) and global average pooling, trading architectural fidelity for the strong transfer-learning prior of ImageNet pretraining, which is better suited to our small-data regime.

Projection block. Following standard practice [3], the 2048-d backbone output is compressed by: $\text{BN}(2048) \rightarrow \text{Dropout}(0.0) \rightarrow \text{Linear}(2048, 512, \text{bias}=\text{False}) \rightarrow \text{BN}(512)$. This produces a 512-d embedding $\mathbf{z} \in \mathbb{R}^{512}$ (unnormalised).

ArcFace head [3]. Let $\hat{\mathbf{z}} = \mathbf{z}/\|\mathbf{z}\|$ be the L2-normalised embedding and $\hat{\mathbf{w}}_k$ the L2-normalised class weight for identity k . Define the angle θ_y between the embedding and the ground-truth class weight. The ArcFace logit is:

$$\ell_y = s \cdot \cos(\theta_y + m), \quad \ell_{k \neq y} = s \cdot \cos \theta_k \quad (1)$$

where $m = 0.2$ (margin, radians) and $s = 30$ (scale) in the tuned configuration used in our experiments; logits are passed through cross-entropy.

AdaFace head [8]. AdaFace augments the ArcFace formulation with an image-quality proxy derived from $\|\mathbf{z}\|$, the feature norm of the unnormalised embedding. A per-batch running mean μ_b and standard deviation σ_b of feature norms are maintained via exponential moving average (EMA momentum $\alpha = 0.01$). The margin is then adapted per sample:

$$g_{\text{margin}} = \text{clip}\left(\frac{\|\mathbf{z}\| - \mu_b}{\sigma_b + \varepsilon} \cdot h, -1, 1\right), \quad (2)$$

$$\ell_y = s \cdot [\cos(\theta_y - m \cdot g_{\text{margin}}) - (m \cdot g_{\text{margin}} + m)] \quad (3)$$

with $h = 0.333$, $m = 0.2$, $s = 30$. The constant additive term $+m$ (Eq. 19 of Kim et al. [8]) is essential: at average quality ($g_{\text{margin}}=0$) the head reduces to CosFace with additive margin m , not to a plain-softmax logit; the limit cases are $g_{\text{margin}}=-1$ (ArcFace, $\cos(\theta_y + m)$), $g_{\text{margin}}=0$ (CosFace, $\cos \theta_y - m$), and $g_{\text{margin}}=1$ (negative angular margin with a shift). Images with below-average feature norms (low quality) thus receive a reduced effective margin, lessening their influence on the decision boundary.

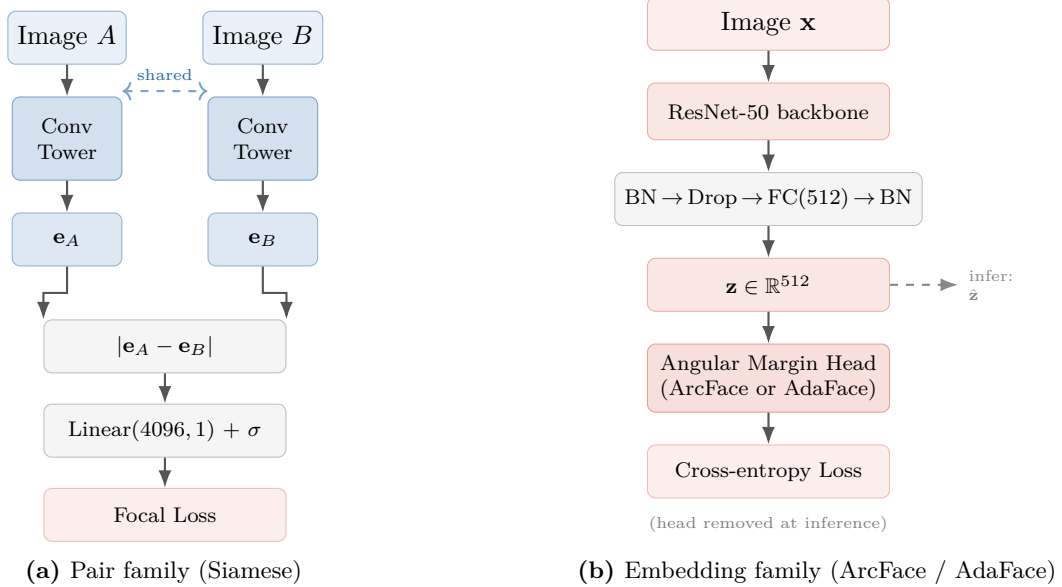


Figure 2: Training topologies for the two model families. *Left:* the Siamese pair-based network uses weight-tied towers and an L1+sigmoid decision head trained with Focal loss. *Right:* the embedding-based network uses a ResNet-50 backbone with a projection block and an angular-margin head. At inference the head is discarded and cosine similarity on L2-normalised embeddings drives verification.

Inference. At test time both ArcFace and AdaFace return the L2-normalised embedding $\hat{z} = z / \|z\|$ without the classification head. Verification uses cosine similarity as in the Siamese case.

Figure 2 summarises both training topologies.

4.3 Hyperparameter tuning

Table 2 summarises the final training configurations. The embedding-model default configuration (SGD, $\eta = 0.1$, $m = 0.5$, $s = 64$) collapsed completely on our small datasets (ROC-AUC = 0.50, Section 6), as the paper-scale learning rate destroyed the ImageNet pretrained backbone before meaningful features were learned. The tuned configuration (Adam, $\eta = 10^{-3}$, $m = 0.2$, $s = 30$) was selected by Optuna [1] hyperparameter search on the validation split.

5 Evaluation Protocol

5.1 Gallery and probe construction

For each test split we designate one image per identity as the *gallery* (the enrolment reference) and treat the remaining images as *probes*. Gallery entries are selected deterministically by sorting by file path and taking the first image; this ensures reproducibility. Identities with fewer than $g + 1$ images (gallery size $g = 1$ plus at least one probe) are excluded from evaluation.

5.2 Verification

Verification asks: *do two images show the same individual?* All $\binom{N_g + N_p}{2}$ unordered pairs are generated from the combined gallery and probe pool; a pair is labelled positive if both images share the same identity. The similarity score for each pair is the cosine similarity of

Table 2: Training hyperparameters.

Hyperparameter	Siamese	ArcFace / AdaFace
Optimiser	Adam	Adam
Learning rate	5×10^{-5}	1×10^{-3}
Weight decay	0.0	5×10^{-4}
LR schedule	constant	cosine + 1 warmup epoch
Max epochs	80	40
Early-stop monitor	Recall@FAR $\leq 10^{-2}$	val ROC-AUC
Early-stop patience	5	7
Mixed precision	bf16	bf16
EMA decay	0.0	0.999
Loss margin (m)	—	0.2
Loss scale (s)	—	30
Focal α/γ	0.75 / 2.0	—

L2-normalised embeddings:

$$s(\mathbf{x}_i, \mathbf{x}_j) = \frac{\hat{\mathbf{z}}_i \cdot \hat{\mathbf{z}}_j}{\max(\|\hat{\mathbf{z}}_i\|, \varepsilon) \max(\|\hat{\mathbf{z}}_j\|, \varepsilon)}, \quad \varepsilon = 10^{-12}. \quad (4)$$

The following metrics are computed:

ROC-AUC. Area under the receiver operating characteristic curve, computed via `sklearn.metrics.roc_auc_score`. Provides a threshold-independent measure of discriminative power.

Youden’s J and optimal threshold. $J = \text{TPR} - \text{FPR}$ (equivalently, sensitivity + specificity -1) [21], maximised over the ROC curve to yield an optimal decision threshold. This is used for F_1 , precision, recall, and specificity computation.

Threshold-selection policy. All operating-point thresholds (Youden’s J, Recall@FAR $\leq 10^{-2}$, and the TAR@FAR budgets) are selected on the *validation* split and then applied unchanged to the test split; the resulting test operating points are reported under the `test_val_threshold/*` prefix. Thresholds re-fit directly on the test ROC curve are retained only as diagnostic upper bounds, never as headline numbers, to avoid optimistic bias from tuning the decision boundary on the evaluation set. The permutation test below likewise uses the validation-derived threshold so that the null distribution reflects a deployable, test-blind operating point.

TAR@FAR. The True Acceptance Rate at a fixed False Acceptance Rate (FAR) budget, defined as the maximum TPR among ROC operating points with $\text{FPR} \leq \text{FAR}_{\text{target}}$. We report TAR at FAR = 10^{-3} and 10^{-4} .

Recall@FAR $\leq 10^{-2}$ (headline metric). With test partitions of at most a few hundred pairs, a FAR budget of 10^{-3} permits fewer than one false match and degenerates to noise. We therefore

adopt $\text{Recall@FAR} \leq 10^{-2}$ as the primary biometric operating metric. This corresponds to the TAR at the most lenient operating point not exceeding a 1% false positive rate, offering a meaningful trade-off between security and usability at the current dataset scale.

5.3 Identification

Identification asks: *given a probe image, which gallery identity does it belong to?* For each probe, cosine similarities to all gallery entries are ranked in descending order. Identification metrics are:

- **Top-1 / Top-5 accuracy:** fraction of probes for which the correct identity is ranked first (or within the top 5).
- **CMC curve:** cumulative match characteristic up to rank $k = 10$, $\text{CMC}[k] =$ fraction of probes with first correct match at rank $\leq k$.
- **mAP:** mean average precision in the information-retrieval sense, averaging the per-probe average precision (AP) over all probes.

5.4 Statistical significance: permutation test

To assess whether observed performance exceeds chance, we run an *inference-mode permutation test*: identity labels in the test manifest are permuted $n = 100$ times; for each permutation, embeddings are re-computed (no retraining) and the target metric is evaluated. The one-sided p-value with continuity correction is:

$$p = \frac{\#\{\text{null} \geq \text{observed}\} + 1}{n + 1}. \quad (5)$$

Significance threshold $\alpha = 0.05$. The inference-mode variant is substantially faster than full-retrain permutation testing while still providing a valid null distribution for the embedding quality.

6 Experiments and Results

6.1 Leaky vs. honest evaluation

We first quantify the impact of the evaluation protocol. A Siamese network was trained on the rousettus random-background dataset once under a *random-pair* split (all identities available in all partitions) and once under the *identity-disjoint* protocol. Results are reported in Table 3.

Table 3: Impact of evaluation protocol. Both rows use the same Siamese architecture on rousettus data; only the splitting strategy differs. The random-pair row is from the original (legacy) random-pair pipeline—the current codebase enforces identity-disjoint splits by construction—while the identity-disjoint row is from the re-run on the fixed code.

Protocol	ROC-AUC	F ₁	Recall
Random-pair (leaky)	0.982	0.987	0.986
Identity-disjoint (correct)	0.858	0.707	0.734

The 12-point drop in ROC-AUC and 28-point drop in F_1 confirm that random-pair splitting substantially overstates real-world performance. All results below use the identity-disjoint protocol exclusively.

6.2 Model family comparison on rousettus

Table 4 presents test-set results for the three model families on the rousettus dataset under both the random-composited and green-screen background conditions (3 unseen test identities in each).

Table 4: Test results on rousettus (identity-disjoint). ArcFace v1 = paper-scale default hyperparameters; ArcFace v2 / AdaFace = tuned hyperparameters. ROC-AUC for *Siamese* is computed from its native pairwise head; for ArcFace/AdaFace it is cosine-similarity verification on the embeddings, so the two are **not directly comparable** (Siamese cosine ROC-AUC, secondary: random 0.946, green 0.959). “—” = not applicable. **Bold** marks the best identification metric per background. All rows except those marked [†] are significant at $p < 0.05$ (permutation test, $n = 100$).

Model	BG	ROC-AUC	F_1	Top-1	mAP	TAR@10 ⁻³
Siamese	random	0.858	0.707	—	—	0.012
Siamese	green	0.669	0.797	—	—	0.001
ArcFace (v2)	random	0.665 [†]	—	0.532	0.731	0.074
ArcFace (v2)	green	0.847	—	0.846	0.915	0.067
AdaFace	random	0.434 [†]	—	0.274	0.560	0.003
AdaFace	green	0.753	—	0.556	0.762	0.006
ArcFace (v1)	random	0.500 [†]	—	0.591	0.757	0.000

On the *random* background, the Siamese network leads decisively (pairwise-head ROC-AUC 0.858, $p < 0.05$), while the tuned embedding models stay at or near chance and are *not* statistically significant (ArcFace 0.665, $p = 1.0$; AdaFace 0.434, $p = 0.78$). On the *green-screen* background, however, the embedding models recover sharply and become significant: ArcFace reaches ROC-AUC 0.847 with Top-1 0.846 and mAP 0.915, exceeding the Siamese network on the identification metrics. This background dependence—rather than identity count alone—is the dominant factor in the embedding models’ behaviour (Section 7). ArcFace v1 with paper-scale hyperparameters still collapses to ROC-AUC = 0.50 (it degenerates to predicting every pair “same”, recall 1.0), confirming that the large learning rate ($\eta = 0.1$) destroys the ImageNet features before any face-specific knowledge is acquired.

6.3 Model family comparison on mauritius

Table 5 extends the comparison to the mauritius species across three background conditions.

The Mauritius results sharpen the picture. On the controlled *green-screen* condition the embedding models are now the strongest: AdaFace attains ROC-AUC 0.902, Top-1 0.916, and mAP 0.955, surpassing both ArcFace and the Siamese network and demonstrating that—with the corrected AdaFace margin, ImageNet normalisation, and matched 112-px evaluation—angular-margin embeddings generalise well to unseen identities even at ≤ 13 training individuals *when the background is coherent*. On the *original* background the embedding models remain solid (ArcFace 0.791, AdaFace 0.731), whereas on *random-composited* backgrounds they fall back toward

Table 5: Test results on mauritius by background condition (identity-disjoint). “Siamese” uses Focal loss; “ArcFace”/“AdaFace” use tuned angular losses. As in Table 4, Siamese ROC-AUC is pairwise-head and the embedding ROC-AUC is cosine-based (not directly comparable; Siamese cosine ROC-AUC, secondary: green 0.891, original 0.927, random 0.820). **Bold** marks the best identification metric per background. [†] = not significant at $p < 0.05$.

Model	BG	ROC-AUC	F ₁	Top-1	mAP	TAR@10 ⁻³
Siamese	green	0.710	0.689	—	—	0.014
Siamese	original	0.834	0.753	—	—	0.005
Siamese	random	0.806	0.686	—	—	0.007
ArcFace	green	0.843	—	0.740	0.870	0.282
AdaFace	green	0.902	—	0.916	0.955	0.383
ArcFace	original	0.791	—	0.576	0.774	0.145
AdaFace	original	0.731	—	0.634	0.785	0.151
ArcFace	random	0.578 [†]	—	0.479	0.697	0.021
AdaFace	random	0.539	—	0.459	0.682	0.006

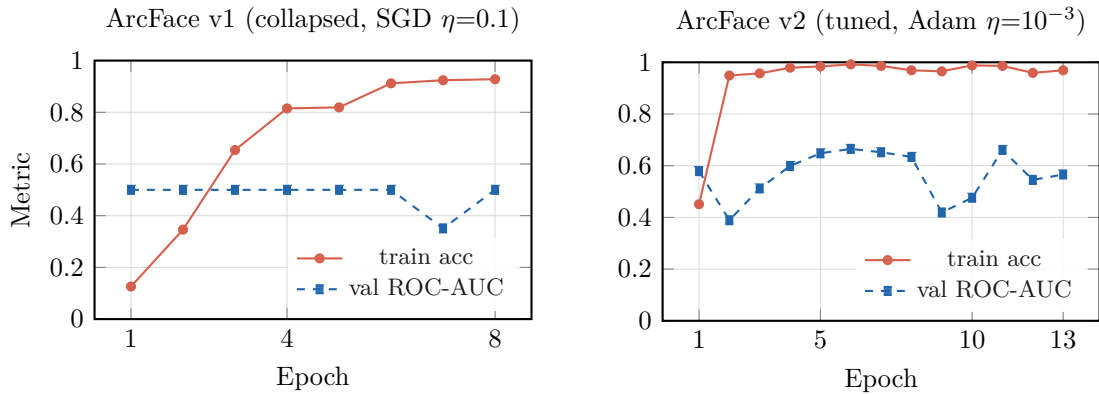


Figure 3: Training dynamics of ArcFace on roussettus (random background). *Left:* the default (paper-scale) SGD configuration fails—train accuracy climbs to ~ 0.93 while val ROC-AUC stays pinned at chance (0.50), the hallmark of a closed-set classifier that fits the training identities but learns no transferable verification geometry. *Right:* the tuned Adam configuration reaches near-perfect train accuracy but val ROC-AUC oscillates and peaks only around 0.66, revealing the generalisation gap to unseen test identities under random-background compositing.

chance (ArcFace 0.578, not significant; AdaFace 0.539). The Siamese network, by contrast, is comparatively stable across all three backgrounds (pairwise-head ROC-AUC 0.71–0.83; cosine ROC-AUC 0.82–0.93) and significant throughout. Background condition, not identity count, is thus the variable that most strongly separates the two families.

6.4 ArcFace hyperparameter sensitivity

Figure 3 illustrates the training dynamics under the two ArcFace configurations.

6.5 Learning curves

Figure 4 shows per-epoch validation curves for all three models on roussettus (random background).

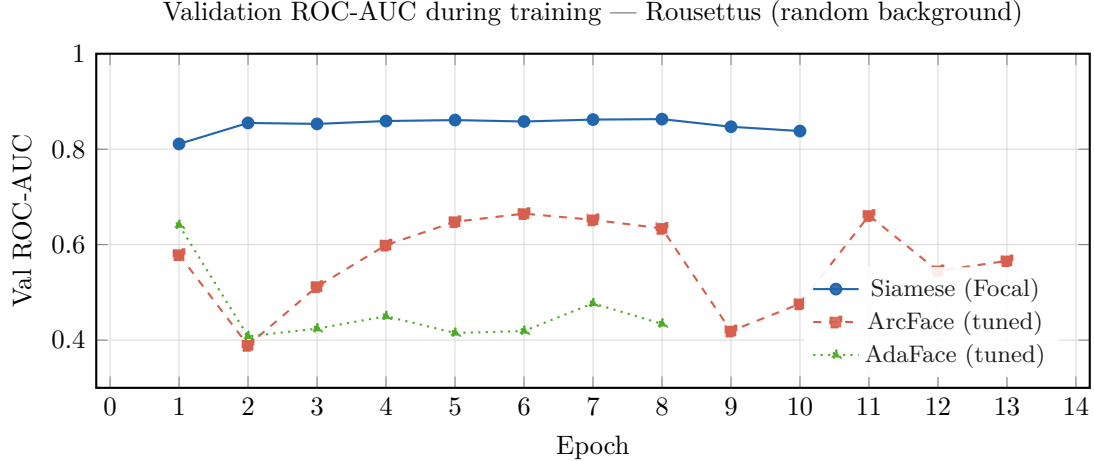


Figure 4: Validation ROC-AUC per training epoch on rousettus (random background). The Siamese network converges quickly and stably to ~ 0.86 . ArcFace and AdaFace show high epoch-to-epoch variance and never stabilise on this random-background condition, with AdaFace decaying toward chance, reflecting the difficulty of learning transferable angular geometry under random-background compositing. Early stopping terminates the embedding runs after 8–13 epochs (best-validation checkpoint restored for test).

6.6 Family and background comparison (summary bar chart)

Figure 5 provides a visual summary of ROC-AUC across all experimental conditions.

6.7 Reproducibility

Table 6 documents the effect of the DataLoader seeding fix on run-to-run reproducibility for ArcFace. Without an explicit `torch.Generator` in the `DataLoader`, the global RNG state (already advanced by model initialisation and EMA setup) causes non-deterministic batch ordering even with `deterministic=True`.

Table 6: Run-to-run reproducibility for ArcFace (rousettus, random BG). Repro-1/2: before DataLoader seed fix; Repro-3/4: after fix.

Run	test ROC-AUC	test Top-1
repro-1 (pre-fix)	0.528	0.326
repro-2 (pre-fix)	0.563	0.405
repro-3 (post-fix)	0.552	0.361
repro-4 (post-fix)	0.552	0.361

Post-fix, the two identical runs (repro-3 and repro-4) produce bit-exact identical metrics, confirming that the fix resolves non-determinism.

7 Discussion

7.1 Two robust families, and one decisive covariate: background

Two findings organise the corrected results. First, the Siamese network is *robust*: it produces statistically significant verification (permutation test, $p < 0.05$) on every species and background, with pairwise-head ROC-AUC in 0.67–0.86 and cosine ROC-AUC in 0.82–0.96. Second,

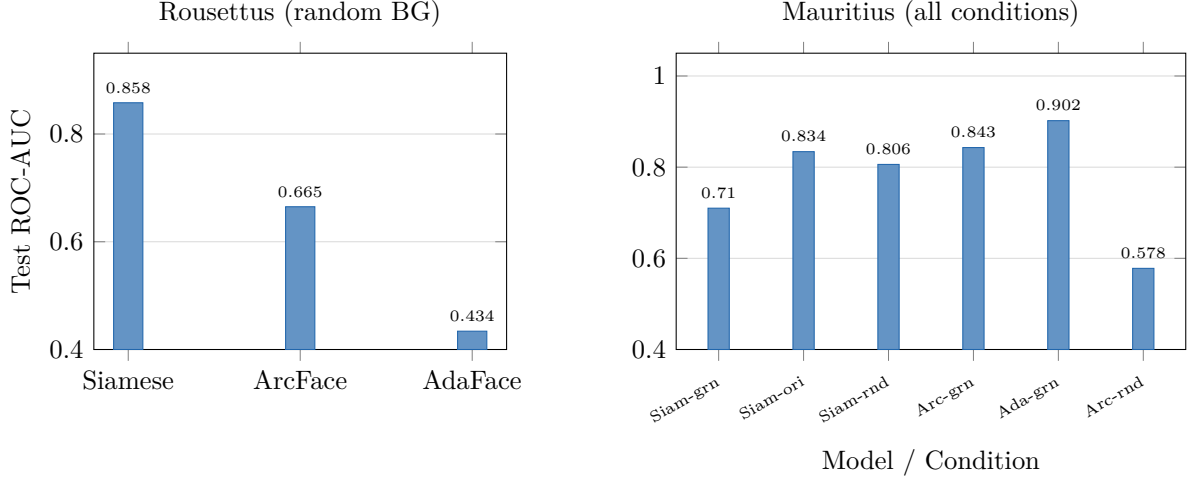


Figure 5: Test ROC-AUC for the major conditions. *Left:* rousettus, random background. *Right:* mauritius across background conditions and model families. “Siam” = Siamese, “Arc” = ArcFace, “Ada” = AdaFace; suffixes grn/ori/rnd indicate green/original/random backgrounds. Siamese bars use the pairwise-head ROC-AUC and the ArcFace/AdaFace bars use cosine-verification ROC-AUC, so heights are not directly comparable across families; the figure is intended to show the within-family background trend, in particular the embedding models’ recovery on green/original and their collapse on random.

the embedding models (ArcFace and AdaFace), once the AdaFace additive margin, ImageNet normalisation, and matched 112-px evaluation are corrected, are *not* uniformly weak—their behaviour is governed almost entirely by the background condition rather than by the number of training identities.

This is the central revision relative to our earlier reading of these experiments. We had attributed the embedding models’ weakness to a “small-identity generalisation wall.” The corrected runs show that with ≤ 13 training identities the very same models reach—and on Mauritius green *exceed*—the Siamese network when the background is coherent (AdaFace Mauritius green: ROC-AUC 0.902, Top-1 0.916, mAP 0.955). The factor that separates success from collapse is therefore not identity count alone but whether the test imagery shares the controlled background of training.

Objective alignment still favours pairs under distribution shift. The structural reason the Siamese network is robust remains valid: it optimises a *pairwise verification* objective that matches the deployment criterion and never binds identities to fixed classifier slots. Angular-margin losses instead solve a *closed-set classification* problem and rely on the learned hypersphere geometry transferring to unseen identities. What the new data clarifies is that this transfer succeeds at small identity counts *when the input distribution is stable*, and fails when random-background compositing injects nuisance variation that the closed-set geometry has not been forced to factor out.

7.2 Background sensitivity, not identity count, drives the gap

Three signals, re-examined across background conditions, converge on background as the governing variable.

The train/test gap is background-dependent, not count-dependent. Training accuracy is near-perfect for the embedding models in *every* condition, so identity count alone cannot explain the test gap. What moves the gap is the background:

Table 7: Embedding-model train accuracy vs. test ROC-AUC on roussettus, by background. Identity count is fixed (6 train identities) across rows; only the background differs. The generalisation gap is large on random compositing and small on green screen.

Model	BG	Train acc	Test ROC-AUC	Gap
ArcFace (tuned)	random	0.969	0.665	−0.304
AdaFace (tuned)	random	0.975	0.434	−0.541
ArcFace (tuned)	green	0.998	0.847	−0.151
AdaFace (tuned)	green	0.998	0.753	−0.245

With the identity count held fixed, moving from random to green backgrounds roughly halves the ArcFace gap ($-0.304 \rightarrow -0.151$) and substantially reduces the AdaFace gap. A pure identity-count bottleneck would not be relieved by changing the background.

Top-1 identification is at chance only on random backgrounds. With 3 unseen test identities, chance Top-1 is 0.33. On random backgrounds the embedding models hover near chance (ArcFace roussettus 0.53, AdaFace 0.27), but on green screen they are far above it (ArcFace roussettus 0.85, Mauritius 0.74; AdaFace Mauritius 0.92). The embedding space *does* acquire transferable structure for novel identities—provided the background does not dominate the signal.

Optimization instability tracks the background too. The validation ROC-AUC curves for ArcFace and AdaFace on random backgrounds (Figure 4) show high epoch-to-epoch variance and, for AdaFace, decay toward chance, whereas the Siamese network converges smoothly. On green/original backgrounds the same losses converge to high, stable validation scores (Section 6). The instability is thus a property of the random-background optimisation landscape rather than an intrinsic small-class pathology.

Cross-species consistency points to background, not species. The pattern replicates across both species, which rules out species-specific causes:

Table 8: Embedding (ArcFace) test ROC-AUC by background and species (tuned). Random-background weakness and green-screen recovery are consistent across both species, isolating background—not species or identity count—as the common cause.

Species	ArcFace random	ArcFace green	Δ
rousettus	0.665	0.847	+0.182
mauritius	0.578	0.843	+0.265

Both species are weak on random compositing and both recover strongly on green screen. Had

identity count been the binding constraint, the green-screen runs—at the same small counts—would not recover.

A softened role for identity count. None of this means identity count is irrelevant. The literature is unambiguous that angular-margin losses improve with many training identities [3, 8], and our random-background runs remain the hardest precisely where nuisance variation compounds limited identity diversity. Our revised reading is more specific: at this scale the *binding* constraint we can demonstrate is background coherence, and the apparent identity-count “wall” in earlier runs was confounded both by the random-background condition and by the AdaFace-margin and evaluation-resolution defects that this study corrects. Disentangling residual identity-count effects from background effects would require a controlled sweep of identity count at fixed background, which we leave to future work.

7.3 Background effects

Across families, coherent backgrounds help, but the effect is far larger for the embedding models than for the Siamese network. On Mauritius the Siamese pairwise-head ROC-AUC varies only modestly with background (0.71–0.83), whereas ArcFace swings from 0.578 (random) to 0.843 (green) and AdaFace from 0.539 to 0.902 (Table 5). Two mechanisms are plausible: (1) random texture compositing introduces visual clutter that competes with facial features in the input field; (2) consistent backgrounds reduce domain shift within the test split. The markedly stronger sensitivity of the angular-margin geometry indicates that, unlike the pairwise objective, it does not automatically learn to factor out background nuisance at this data scale—an actionable finding, since background standardisation at capture time is cheap relative to collecting many more identities.

7.4 The leaky-vs-honest gap

The 28-point F_1 gap between the random-pair protocol and the identity-disjoint protocol (Table 3) is larger than typical published figures for animal re-identification. Practitioners are encouraged to audit their evaluation protocols: if the same individual appears in both train and test, reported numbers are not measuring deployed-scenario performance.

7.5 Limitations

Single-seed evaluation. All reported numbers come from single training runs. The standard deviation of test metrics across seeds is unknown and likely non-trivial for the embedding models given the instability documented in Section 7.2: a repeated AdaFace run on rousettus random produced test ROC-AUC 0.671 versus the reported 0.434 from an identical configuration, underscoring that the random-background embedding numbers in particular should be read as single-sample point estimates with wide implied uncertainty.

No external validation set. The test identities were drawn from the same colony as the training identities. Generalisation to bats from different colonies or imaging setups is untested.

No ablation on backbone pretraining. All embedding models used ImageNet-pretrained ResNet-50 weights. Whether domain-adaptive pretraining on a larger bat-face corpus would lower the identity-count threshold is an open question.

8 Conclusion

We presented an open-set individual bat face recognition system using a strict identity-disjoint evaluation protocol. Our comparative experiments across three model families, two species, and three background conditions lead to three conclusions:

1. **Identity-disjoint evaluation is non-negotiable.** The same architecture can appear to achieve $F_1 \approx 0.99$ under random-pair splitting and $F_1 \approx 0.71$ under identity-disjoint evaluation on the same images. Any reported result on animal re-identification should explicitly state whether splits are identity-disjoint.
2. **Pair-based models are the safe default, but corrected embedding models are competitive—and the deciding factor is background, not identity count.** The Siamese network is the only family that is statistically significant across every species and background. However, once the AdaFace additive margin, ImageNet normalisation, and matched 112-px evaluation are corrected, ArcFace and AdaFace match or exceed the Siamese network on coherent (green/original) backgrounds even at ≤ 13 training identities—AdaFace reaches ROC-AUC 0.902 and Top-1 0.916 on Mauritius green. Their failures concentrate on random-composited backgrounds, where they regress toward chance and lose significance. We therefore frame the embedding-model shortfall as a *background-sensitivity* effect rather than a fixed small-identity wall; identity count likely still matters, but at this scale background coherence is the constraint we can demonstrate.
3. **Controlled backgrounds are a high-leverage intervention, especially for embedding models.** Moving from random to controlled backgrounds lifts embedding-model ROC-AUC by up to ~ 27 points (Mauritius: ArcFace $0.578 \rightarrow 0.843$, AdaFace $0.539 \rightarrow 0.902$) and improves the Siamese cosine ROC-AUC as well, indicating that background standardisation at capture time is cheaper and more impactful than collecting many additional identities.

Reproducibility. All experiments are fully reproducible from the publicly released configuration files and MLflow run IDs cited throughout this paper. The single-command pipeline `uv run bat-cli train -experiment <name>` regenerates the complete training, evaluation, and reporting artefacts from scratch.

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A Software Architecture

A.1 Workspace layout

The codebase is structured as a **monorepo of 13 single-purpose Python packages**, all sharing a `uv` workspace [20] under the root directory. Table 9 lists all packages and their roles.

A.2 Dependency graph and invariants

The workspace enforces a strict **no-upward-dependency** rule: a package may only import from `bat_core` or from packages strictly lower in the dependency order. Figure 6 shows the dependency graph.

Key invariants maintained by the architecture:

- **bat_core stays import-light.** It type-annotates `torch.Tensor` only under `TYPE_CHECKING`; runtime import never triggers PyTorch loading.
- **Trainers are loops, not god-classes.** Data wiring belongs to `bat_data`; MLflow logging to `bat_tracking`; eval to `bat_evaluation`. The trainers only own the optimiser, scheduler, AMP, EMA, checkpointing, and callbacks.

Table 9: The 13 packages of the bat face recognition workspace.

Package	Responsibility
<code>bat_core</code>	Types, Protocols, dataclass definitions; no I/O, no torch at import time
<code>bat_data</code>	Manifest management, identity-disjoint splitter, <code>BatDataset</code> , pair miners
<code>bat_preprocessing</code>	YOLO segmentation/pose, frame extraction, background generation, quality scoring
<code>bat_models</code>	Siamese, ArcFace, AdaFace <code>nn.Modules</code> and their backbone/head sub-modules
<code>bat_losses</code>	BCE, Focal, Triplet (pair); ArcFace, AdaFace, CosFace, SubCenter (embedding)
<code>bat_training</code>	<code>PairTrainer</code> and <code>EmbeddingTrainer</code> training loops (no eval, no MLflow)
<code>bat_evaluation</code>	Verification and identification protocols; dataclass-only outputs
<code>bat_interpretability</code>	Saliency, GradCAM, embedding projection (t-SNE/UMAP) adapters
<code>bat_stats</code>	Permutation tests, experiment-aware visualiser, naming helpers
<code>bat_tracking</code>	MLflow façade with sectioned metrics and hyperparameter allow-list
<code>bat_reporting</code>	Unified post-training PDF (reportlab) and run-comparison HTML
<code>bat_sweeps</code>	Optuna val-only hyperparameter search (test split never touched)
<code>bat_cli</code>	Single entry-point CLI (<code>bat-cli</code>) with InquirerPy interactive picker

- **Test split is sacred.** The Optuna sweep (`bat_sweeps`) accesses only validation metrics; the objective function never calls `trainer.test`. This is enforced by a unit-test mock.
- **No CSV output.** All evaluation results are returned as typed dataclasses and rendered directly to the unified PDF.
- **Consistent naming.** All plot titles, filenames, and axis labels route through `bat_stats.naming.experiment_name`, which encodes the species/background/model/loss into every output for unambiguous traceability.

A.3 Automated post-training pipeline

A single `bat-cli train -experiment name` command executes the full post-training workflow depicted in Figure 7.

A.4 PDF report structure

The unified post-training PDF generated by `bat_reporting` contains eight sections: title page, training curves, confusion matrix, ROC curve, identification metrics table, saliency composite, t-SNE/UMAP projection, and permutation test summary. It provides a single self-contained artefact for each experimental run, uploaded to MLflow alongside the model checkpoint.

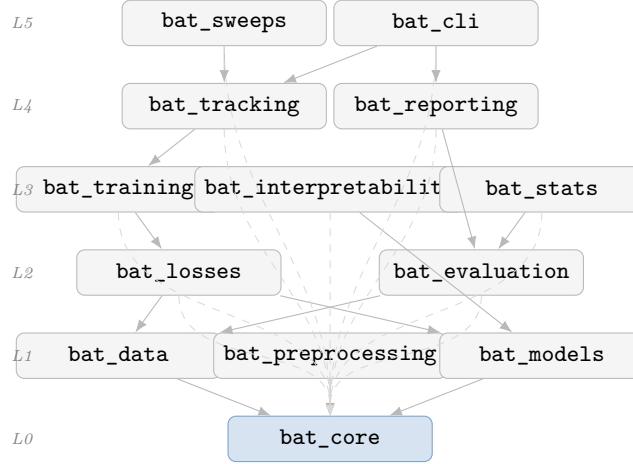


Figure 6: Package dependency graph (layers L0–L5). Solid arrows show the primary inter-layer dependencies. Dashed lines indicate that every package ultimately imports `bat_core` (L0, blue). `bat_core` itself imports nothing from the workspace. The CLI at L5 orchestrates the full stack.

B Technology Stack

B.1 Core framework

All experiments are implemented in **Python 3.11** using **PyTorch** [14] as the deep learning backend. The workspace is managed by `uv`, a fast Python package installer and resolver that replaces `pip/conda` for dependency management. A single `uv sync -frozen` command restores the full pinned environment from `uv.lock`.

B.2 Model and training infrastructure

- **PyTorch** [14]: the primary deep learning framework. All model code uses native `nn.Module` with standard training loops.
- **Torchvision**: provides the ResNet-50 backbone with ImageNet1K-V2 pretrained weights (`ResNet50_Weights.DEFAULT`).
- **HuggingFace Accelerate** [5]: provides device management, AMP (bfloat16) and gradient accumulation; the trainer integrates an `Accelerator` instance for device-agnostic mixed-precision training.
- **EMA (Exponential Moving Average)**: the embedding trainers maintain an EMA copy of model weights ($\text{decay} = 0.999$) using a lightweight in-house implementation; the EMA model is used for all validation and test inference.
- **Early stopping**: implemented with configurable patience and minimum delta; the monitored metric is $\text{Recall@FAR} \leq 10^{-2}$ for pair-based models and val/ROC-AUC for embedding models.

B.3 Configuration management

Experiments are configured via **Hydra** [20], a framework for hierarchical configuration composition. Each run composes one YAML file from each of seven configuration groups:

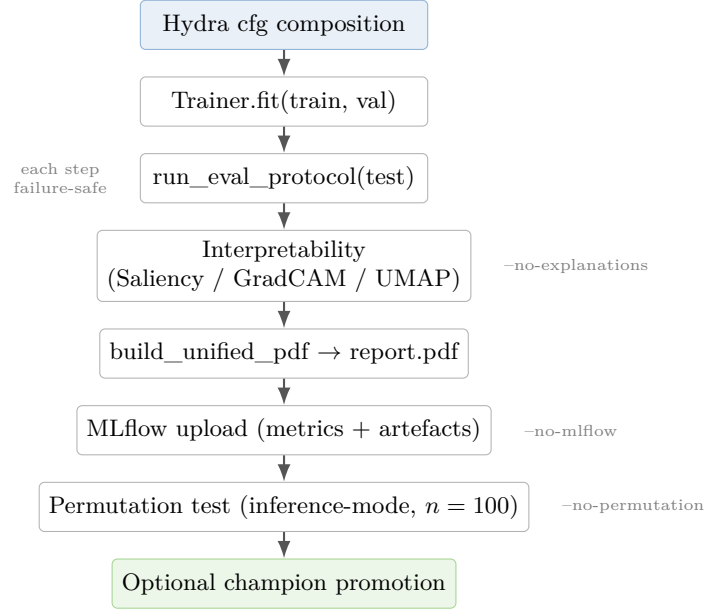


Figure 7: The automated post-training pipeline executed by `bat-cli train`. Each step is wrapped in a safety handler; a failure logs a warning but downstream steps still run. Skip flags (shown in grey) allow selective execution.

`data / model / loss / trainer / evaluation / preprocessing / sweep`

Pre-composed experiment stacks are stored in `configs/experiment/` (10 files covering all species×background×model combinations). The global seed (default 42) is propagated to all random operations.

B.4 Experiment tracking

MLflow [22] is used for experiment tracking. The `bat_tracking` façade logs:

- All training hyperparameters (model, loss, trainer, data params from an allow-list to avoid Hydra internals).
- Per-epoch train/val metrics including loss, accuracy, F_1 , ROC-AUC, TAR@FAR values, and Recall@FAR $\leq 10^{-2}$.
- Test-set metrics after training.
- Artefacts: the unified PDF report, saliency images, and permutation test outputs.

The local MLflow tracking server stores data in `mlruns/` at the repository root. Run IDs (quoted throughout Section 6) uniquely identify each experimental result.

B.5 Hyperparameter optimisation

Optuna [1] provides hyperparameter search over validation metrics only. The search space for ArcFace covers: margin $m \in [0.3, 0.6]$, scale $s \in [32, 96]$, and $\text{lr} \in \log\text{-unif}[10^{-3}, 3 \times 10^{-1}]$, over 20 trials. The Siamese recall sweep covers Focal $\alpha \in [0.5, 0.95]$, $\gamma \in [1, 4]$, and $\text{lr} \in \log\text{-unif}[10^{-5}, 2 \times 10^{-4}]$, over 30 trials. Trial state is persisted to `optuna_studies.db` (SQLite). The test split is never accessed during sweeps (enforced by a unit test mock).

B.6 Preprocessing

Face images are extracted from raw video by the **YOLO**-based pipeline [7, 16]:

1. **YOLOPoseEstimator** (`face_pose.pt`) detects facial keypoints.
2. **FaceAligner** (NumPy/OpenCV) rotates the image so that the two primary keypoints are horizontal, standardising pose.
3. **YOLOSegmenter** (`face_seg.pt`) produces a binary face mask.
4. Square crop around the mask bounding box with outer margin ratio 0.1; canvas padding ensures out-of-bounds crops remain valid.
5. Optional **BackgroundGenerator** replaces or composites the background.
6. Resize to target resolution and normalise to $[0, 1]$.

Predictions are cached (512 MB in-memory LRU cache) to avoid redundant YOLO inference during repeated runs. Frame quality is scored by Laplacian variance (OpenCV) and stored in the manifest.

B.7 Interpretability and reporting

Post-training interpretability is provided by three adapters in `bat_interpretability`:

- **SiameseSaliencyAdapter**: gradient-based input saliency for the pair network.
- **GradCAMAdapter**: class activation maps [6] for embedding models, highlighting regions most discriminative for a given identity.
- **EmbeddingProjectionAdapter**: t-SNE and UMAP projections of test-set embeddings, coloured by identity, providing a 2-D snapshot of the embedding space geometry. To keep every identity visible while bounding plot density, embeddings are sampled with a *max-samples-per-identity* policy (default 8) rather than a single global cap, and each legend entry reports the number of projected samples for that identity, e.g. `id_7 (n=5)`.

The unified PDF report is assembled by **reportlab** (pure-Python PDF generation) and uploaded to MLflow as a run artefact.

B.8 CLI surface

Table 10 lists the complete `bat-cli` command surface.

Invoked without arguments, `bat-cli` presents an interactive picker (powered by `InquirerPy`) from which any action can be selected.

B.9 Quality assurance

The sole quality gate is `pre-commit` (no CI), configured with:

- `ruff` for linting and import sorting.
- `mypy` for incremental static type checking.

Table 10: bat-cli command surface.

Command	Purpose
<code>train</code>	Compose config → train → test eval → interpretability → PDF → MLflow → permutation test → optional promote
<code>sweep</code>	Optuna search on validation split
<code>evaluate</code>	Re-run test evaluation against a saved checkpoint
<code>permutation-test</code>	Inference-mode permutation test on a configured run
<code>compare</code>	Side-by-side MLflow run comparison HTML
<code>compare-champion</code>	Same report, left column auto-resolved via <code>bat_tracking.get_champion</code>
<code>promote</code>	Promote a registered model if criterion beats incumbent
<code>build-manifest</code>	Walk a directory tree → split + write manifest CSV

- Standard hooks (trailing whitespace, end-of-file, YAML/TOML syntax).

Unit tests are co-located with packages and run via `uv run pytest packages/`. The test suite covers the identity-disjoint splitter (including the mock that confirms the test split is never accessed by sweeps), manifest hash integrity, metric computations, and named edge cases in the verification/identification protocols.